FORMULAE LIST

Volume of a sphere: $V = \frac{4}{3}\pi r^3$ Volume of a cone: $V = \frac{1}{3}\pi r^2 h$ Volume of a pyramid: $V = \frac{1}{3}Ah$

1. Expand and simplify $(2x-1)(x^2-5x+3)$. (3)

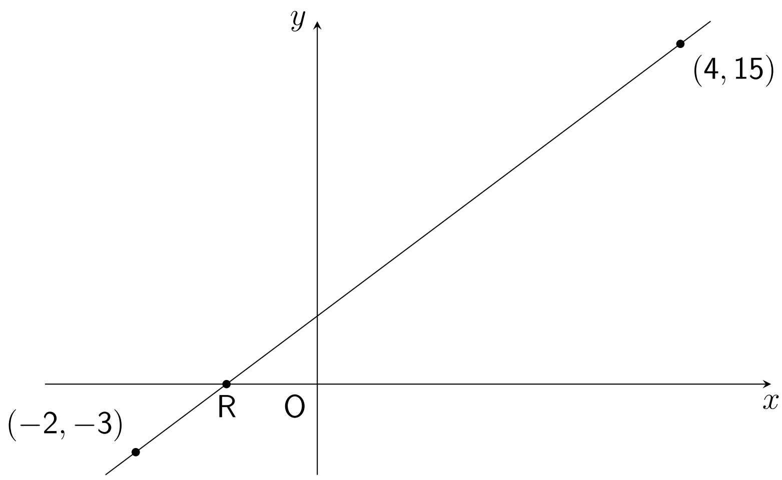
2. Calculate $\dfrac{5}{6} \div 2\dfrac{1}{4}$. (2)

3. Make r the subject of $t = pr^3 - q$. (3)

4. Solve the equation $x^2 - 7x + 10 = 0$. (2)

5. Solve the inequation $1 - 2(3x+4) < 5(x-8)$. (3)

6. (a) Determine the equation of the straight line passing through (−2,−3) and (4,15), shown below. (3)



(b) Using the equation obtained from part (a), find the coordinates of point R, where the line intercepts the x-axis. (2)

Show clear working.

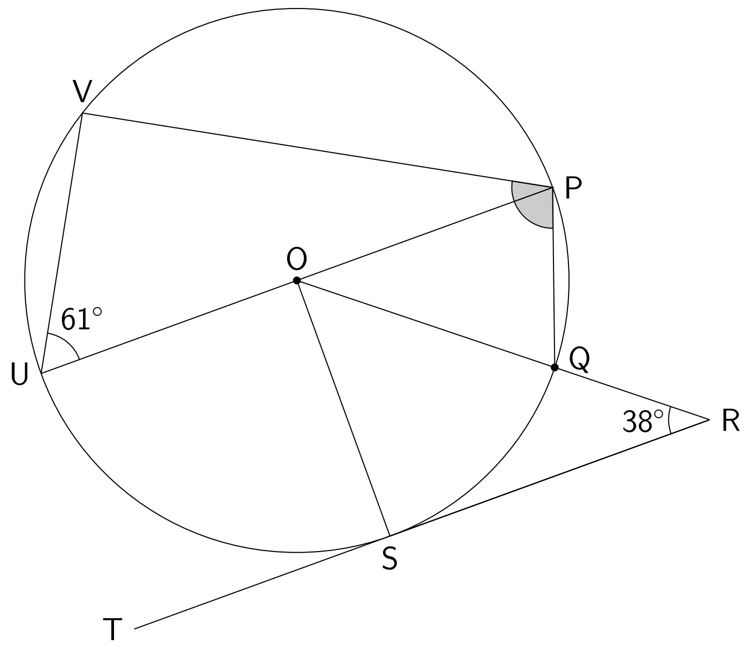
7. Factorise:

   (a) $3x^2 - 9x + 6$ (2)

   (b) $9x^2 - 49$ (2)

8. In the diagram below line TR is a tangent to the circle with centre O at point S. (3)

   UP is a straight line and is parallel to TR.

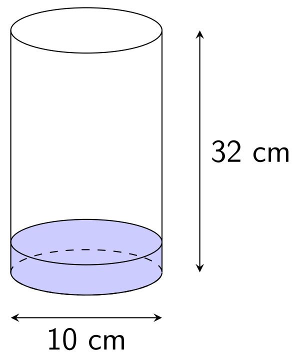


   Find the size of shaded angle VPQ.

9. Simplify $\dfrac{k^4 \times k^5}{\left(k^{-1}\right)^3}$. (3)

10. The cylinder below is filled with water to **one eighth** of its height.

    Its diameter is 10 centimetres and its height is 32 centimetres.



    Calculate the volume of the water. Use $\pi = \mathbf{3.14}$.

END OF PRACTICE EXAM C PAPER 1

HINTS: Practice Exam C Paper 1

1. Check **powers and negatives** carefully.
2. Rewrite the division as a **multiplication** - but what needs to be done **first**?
3. Leave dealing with the **cube** until **last** here.
4. To solve a **quadratic equation**, try **factorising**...
5. First **expand the brackets** carefully, and **simplify** where possible.
6. What is the formula for the **gradient** of a line between two points? What, next, is the **equation for a straight line**? For part (b), use your equation from part (a) - what coordinate do we **already know** for R?
7. The three types of factorising covered in Form III are: **common factors**, **difference of two squares** and **quadratic trinomials / double brackets**. Look for each, in that order.
8. Look for any **right-angles**, and any **isosceles triangles**.
9. Remember the **three laws for indices** covered this year.
10. The formula for the volume of a **cylinder** is similar to the one for a *cone* given in the formula sheet. Note that only a **fraction of the cylinder** contains water.

ANSWERS: Practice Exam C Paper 1

1. $2x^3 - 11x^2 + 11x - 3$
2. $\dfrac{10}{27}$
3. $r = \sqrt[3]{\dfrac{t+q}{p}}$
4. $x = 2, x = 5$
5. $3 < x$ **or** $x > 3$
6. (a) $y = 3x + 3$

   (b) $x = -1$
7. (a) $3(x-1)(x-2)$

   (b) $(3x+7)(3x-7)$
8. $\angle VPQ = 100°$
9. k^{12}
10. 314cm^3